AQ3.3: Activity Questions 3 - Not Graded

This assignment will not be graded and is only for practice.
Level 1
1 point
Which of the following vectors can be written as the linear combination of vectors $(2, 3)$(2, 3) and $(1, 2)$(1, 2) in R^2 R2 with usual addition and scalar multiplication?

- [ ] $(1, 0)$(1, 0)
- [ ] $(0, 1)$(0, 1)
- [ ] $(0, 0)$(0, 0)
- [ ] Any vector in R^2 R2

No, the answer is incorrect.
Score: 0
Accepted Answers:
$(1, 0)$(1, 0)
$(0, 1)$(0, 1)
$(0, 0)$(0, 0)
Any vector in R^2 R2

1 point
Suppose the set $\{(3, 7, 4), v\}$\{(3, 7, 4), v\} is a linearly dependent set in the vector space R^3 R3 with usual addition and scalar multiplication. Which of the following vectors are possible as a candidate for v v?

- [ ] $(0, 0, 0)$(0, 0, 0)
- [ ] $(5, \frac{35}{3}, \frac{20}{3})$(5, 335, 320)
- [ ] $(1, 0, 0)$(1, 0, 0)
- [ ] $(6, 14, -8)$(6, 14, −8)

No, the answer is incorrect.
Score: 0
Accepted Answers:
$(0, 0, 0)$(0, 0, 0)
$(5, \frac{35}{3}, \frac{20}{3})$(5, 335, 320)

1 point
Suppose the set $\{(2, 3, 0), (0, 1, 0), v\}$\{(2, 3, 0), (0, 1, 0), v\} is a linearly dependent set in the vector space R^3 R3 with usual addition and scalar multiplication. Which of the following vectors are possible as a candidate for vv?

- [ ] $(2, 3, 1)$(2, 3, 1)
- [ ] $(1, 2, 0)$(1, 2, 0)
- [ ]

$(1, 3, 0)$(1, 3, 0)

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$(0, 1, 1)$(0, 1, 1)

☐

$(\pi, e, 0)$(π, e, 0)

No, the answer is incorrect.
Score: 0
Accepted Answers:

$(1, 2, 0)$(1, 2, 0)

$(1, 3, 0)$(1, 3, 0)

$(\pi, e, 0)$(π, e, 0)

If $(3, 4) = \alpha(1, 2) + \beta(1, 3)$(3, 4) = α(1, 2) + β(1, 3) in the vector space R^2 R2 with usual addition and scalar multiplication, then find the value of $\alpha + 3\beta$.α + 3β.

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) -1

1 point

If (9, 3, 1) is a linear combination of the vectors $(1, 1, 1), (1, -1, 1)$ and (x, y, z)(1, 1, 1), (1, −1, 1) and (x, y, z) as $2(1, 1, 1) + 3(1, -1, 1) + 4(x, y, z) = (9, 3, 1)$ 2(1, 1, 1) + 3(1, −1, 1) + 4(x, y, z) = (9, 3, 1), then find the value of $x + y + 2z$x + y + 2z.

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 0

1 point

1 point

Let VV be a vector space and v_1, v_2, v_3 and $v_4 \in V$v1, v2, v3 and v4 ∈ V . If v_1v1 is a linear combination of $v_i, i = 2, 3, 4$ $i.e.$ $v_1 = av_2 + bv_3 + cv_4$vi, i = 2, 3, 4 i.e. v1 = av2 + bv3 + cv4 then

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v_2v2 is a linear combination of $v_i, i = 1, 3, 4$vi, i = 1, 3, 4.

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v_3v3 is a linear combination of $v_i, i = 1, 4$vi, i = 1, 4.

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v_2v2 is a linear combination of v_2v2.

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v_4v4 is a linear combination of v_4v4.

No, the answer is incorrect.
Score: 0
Accepted Answers:

v_2v2 is a linear combination of v_2v2.

v_4v4 is a linear combination of v_4v4.

1 point

Consider three vectors $v_1 = (1, 0, 0), v_2 = (1, 1, 0), v_3 = (1, 1, 1)$v1 = (1, 0, 0), v2 = (1, 1, 0), v3 = (1, 1, 1) in the vector space R^3 R3, with usual addition and scalar multiplication. Which of the following sets is (are) true?

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$\{v_1, v_2, v_3\}$ is a linearly dependent set.

- [ ] $\{v_1 + v_2, v_1 + v_3, v_3\}$ is a linearly dependent set.
- [ ] $\{v_2 - v_1, v_1, v_2\}$ is a linearly dependent set.
- [ ] $\{v_1, v_2 - v_1, v_3 - v_2, v_3\}$ is a linearly dependent set.

No, the answer is incorrect.
Score: 0
Accepted Answers:

$\{v_2 - v_1, v_1, v_2\}$ is a linearly dependent set.
$\{v_1, v_2 - v_1, v_3 - v_2, v_3\}$ is a linearly dependent set.

Level - 2

1 point
Let S be a subset of R^3 which is linearly dependent. Which of the following options are true?

- [ ] $S \cup \{(1,0,0)\}$ must be linearly dependent.
- [ ] $S \cup \{v\}$ must be linearly dependent for any $v \in R^3$.
- [ ] $S \setminus \{v\}$ must be linearly dependent for any $v \in R^3$.
- [ ] There may exist some $v \in S$, such that $S \setminus \{v\}$ is still linearly dependent.

No, the answer is incorrect.
Score: 0
Accepted Answers:

$S \cup \{(1,0,0)\}$ must be linearly dependent.
$S \cup \{v\}$ must be linearly dependent for any $v \in R^3$.
There may exist some $v \in S$, such that $S \setminus \{v\}$ is still linearly dependent.

$M_{3\times 3}(R)$ denotes the vector space consisting of real square matrices of order 3 with usual matrix addition and scalar multiplication. Consider the following matrices:

$$M_1 = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}, M_2 = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix}, M_3 = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & -1 \\ 1 & 2 & 1 \end{bmatrix}, M_4 = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 2 \\ 1 & -1 & 1 \end{bmatrix}$$

Use the above information to answer questions 9 and 10.

1 point
Which of the following options are correct?

- [ ] $M_1 - M_2 = 0$, where 0 denotes the zero matrix of order 3.

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$2M_1 + M_2 = M_3$ 2M1 + M2 = M3
☐
$2M_1 - M_2 - M_3 = 0$ 2M1 − M2 − M3 = 0 , where 0 0 denotes the zero matrix of order 3.
☐
$2M_2 - M_1 - M_4 = 0$ 2M2 − M1 − M4 = 0, where 0 0 denotes the zero matrix of order 3.
☐
The tuple (a, b, c)(a, b, c) is unique for which $aM_1 + bM_3 + cM_4 = 0$ aM1 + bM3 + cM4 = 0 holds, where 0 0 denotes the zero matrix of order 3.
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The pair (a, b)(a, b) is unique for which $aM_1 + bM_2 = 0$ aM1 + bM2 = 0 holds, where 0 0 denotes the zero matrix of order 3.

No, the answer is incorrect.
Score: 0
Accepted Answers:
$2M_1 - M_2 - M_3 = 0$ 2M1 − M2 − M3 = 0 , where 0 0 denotes the zero matrix of order 3.
$2M_2 - M_1 - M_4 = 0$ 2M2 − M1 − M4 = 0, where 0 0 denotes the zero matrix of order 3.
The pair (a, b)(a, b) is unique for which $aM_1 + bM_2 = 0$ aM1 + bM2 = 0 holds, where 0 0 denotes the zero matrix of order 3.

1 point
Which of the following options is(are) true?
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$\{M_1, M_2\}$ {M1, M2} is a linearly dependent set.
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$\{M_1, M_2, M_3\}$ {M1, M2, M3} is a linearly dependent set.
☐
$\{M_1, M_2, M_4\}$ {M1, M2, M4} is a linearly dependent set.
☐
$\{M_1, M_3, M_4\}$ {M1, M3, M4} is a linearly dependent set.

No, the answer is incorrect.
Score: 0
Accepted Answers:
$\{M_1, M_2, M_3\}$ {M1, M2, M3} is a linearly dependent set.
$\{M_1, M_2, M_4\}$ {M1, M2, M4} is a linearly dependent set.
$\{M_1, M_3, M_4\}$ {M1, M3, M4} is a linearly dependent set.

Check Answers
Your score is: 0/10